Assessment Book

Physics | Grade 9-10 | Laws of Motion

Period 1: Introduction to Newton's Laws of Motion

MCQs: What is Newton's First Law of Motion? (A) An object will accelerate in the direction of the net force (B) An object at rest stays at rest, and an object in motion stays in motion unless acted upon by an unbalanced force (C) For every action, there is an equal and opposite reaction (D) An object will stop moving if it encounters friction.; What does inertia refer to? (A) The speed of an object (B) The tendency of an object to resist changes in its state of motion (C) The weight of an object (D) The temperature of an object.

Short Answer: What do you think would happen to a ball rolling on a smooth surface if no one touches it? Why?; Can someone summarize what inertia is in their own words?; Give me an example of an object demonstrating inertia in your daily life.

Long Answer: In your own words, describe how inertia affects our everyday experiences with motion. Provide one concrete example.

Numerical:

Answer Key: question: What is Newton's First Law of Motion?

answer: B

question: What does inertia refer to?

answer: B

question: What do you think would happen to a ball rolling on a smooth surface if no one touches it? Why?

answer: The ball would keep rolling indefinitely due to inertia, as there is no unbalanced force acting on it.

question: Can someone summarize what inertia is in their own words?

answer: Inertia is the tendency of an object to resist changes in its state of motion.

question: Give me an example of an object demonstrating inertia in your daily life.

answer: An example could be a passenger lurching forward in a bus when it suddenly stops.

question: In your own words, describe how inertia affects our everyday experiences with motion. Provide one concrete example.

answer: Inertia affects our everyday experiences by causing objects to stay in their state of motion. For example, when driving, if the car suddenly stops, passengers feel a jolt forward because their bodies want to continue moving due to inertia.

Rubric: Responses for short answer questions are evaluated based on clarity, completeness, and relevance to the concept of inertia. Long answer responses are assessed based on understanding of inertia, depth of explanation, and the quality of the provided example. Detailed and insightful responses earn full points, while vague or incomplete answers will receive partial points.

Period 2: Newton's First Law of Motion

MCQs: Which of the following best describes Newton's First Law of Motion? (A) An object at rest stays at rest, (B) An object in motion stays in motion, (C) An object will remain unchanged unless acted upon by an unbalanced force, (D) All of the above.; What does inertia refer to in the context of Newton's First Law? (A) The tendency of an object to rest, (B) The tendency of an object to resist changes in its state of motion, (C) The speed of an object, (D) The force acting on an object.

Short Answer: What happens to a ball if no external force is acting on it?; Define unbalanced force in your own words.

Long Answer: Explain Newton's First Law of Motion and provide an example from your daily life that illustrates this principle.

Numerical:

Answer Key: mcqs: D

B

short_answer: The ball will remain at rest if it is not moving, or continue moving at a constant velocity if it is already in motion.

An unbalanced force is a force that causes a change in the motion of an object, meaning that there are forces acting on the object that are not equal and opposite.

long_answer: Newton's First Law of Motion states that an object at rest will stay at rest and an object in motion will stay in motion at a constant velocity unless acted upon by an unbalanced force. An example could be: when I am riding in a car and the driver suddenly brakes, I tend to lurch forward because my body wants to continue moving at the same speed.

Rubric: Responses should be graded on a scale of 1 to 5. For multiple choice questions, 1 point for each correct answer. For short answer questions, 1 point for correct explanation and clarity. For long answer questions, 1 point for explaining Newton's First Law, 1 point for clarity, and 1 point for providing a relevant and clear example.

Period 3: Newton's Second Law of Motion

MCQs: What does the formula F = m * a represent? (A) Mass times acceleration, (B) Force equals mass times acceleration, (C) Acceleration equals force divided by mass, (D) None of the above; How does increasing the force applied to an object affect its acceleration? (A) Increases acceleration, (B) Decreases acceleration, (C) No effect, (D) Changes direction but not speed; If the mass of an object is doubled while the force remains the same, what happens to its acceleration? (A) Doubles, (B) Halves, (C) Stays the same, (D) Quadruples

Short Answer: Describe the forces acting on a soccer ball when a player kicks it.; How does changing the mass of an object affect its acceleration when the same force is applied?

Long Answer: Explain in one or two sentences how Newton's Second Law can help us understand the importance of seat belts in cars.; Provide a real-life example of Newton's Second Law and explain it, including the forces involved and how they relate to mass and acceleration.

Numerical:

Answer Key: 1. B; 2. A; 3. B; Short answers should describe forces such as applied force and gravitational force; Long answers should mention how seat belts help in managing acceleration in a crash and examples like a car vs. a truck showing mass differences.

Rubric: For short answers, 1 point for identifying forces, 2 points for clarity. For long answers, 1 point for relevance to Newton's Second Law, 2 points for including forces, and 2 points for explanation of mass and acceleration.

Period 4: Newton's Third Law of Motion

MCQs: What does Newton's Third Law of Motion state? (A) An object at rest will stay at rest, (B) The acceleration of an object depends on the net force acting on it, (C) For every action, there is an equal and opposite reaction, (D) Energy cannot be created or destroyed.; In which scenario is Newton's Third Law of Motion clearly demonstrated? (A) A book resting on a table, (B) A soccer ball being kicked, (C) An object falling freely under gravity, (D) A plant growing towards sunlight.

Short Answer: Provide an example of an action-reaction pair from everyday life.; Explain how Newton's Third Law connects with the other laws of motion we've studied.

Long Answer: In your own words, explain how Newton's Third Law of Motion applies to a swimmer. What is the action, and what is the reaction?

Numerical:

Answer Key: 1. C; 2. B; Short Answer: Any reasonable example like 'pushing off the wall in a pool (action) and moving forward (reaction)'; Connection explaining that Newton's Third Law works with the First and Second Laws governing force and motion; Long Answer: The swimmer pushes the water backward (action) which propels them forward (reaction).

Rubric: MCQs: 1 point each; Short Answer: 2 points (1 for example, 1 for clarity); Long Answer: 5 points (2 for explanation of action, 2 for reaction, 1 for clarity). Total: 10 points.

Period 5: Applications of Newton's Laws

MCQs: 1. What does Newton's First Law of Motion describe? (A) Action-reaction pairs (B) Mass times acceleration (C) Inertia (D) Force equals mass times acceleration; 2. Which of the following examples best illustrates Newton's Second Law? (A) A car not moving until the driver presses the gas pedal (B) A rocket launching into space (C) A ball rolling on a flat surface (D) A person jumping off a diving board; 3. In which situation is Newton's Third Law illustrated? (A) A car accelerating down a highway (B) A book resting on a table (C) A swimmer pushing against the water while swimming (D) A bicycle coasting down a hill

Short Answer: Explain how inertia is observed in a real-life situation.; Write the formula for Newton's Second Law.; Provide an example of an action-reaction pair from your daily life.

Long Answer: Describe one way Newton's laws of motion are applied in engineering or mechanics.; Discuss the significance of understanding Newton's laws for engineers and provide a specific example from your knowledge or research.

Numerical:

Answer Key: 1. C, 2. B, 3. C; Short Answer: Various acceptable answers illustrating inertia, $F = m * a$, and examples of action-reaction pairs; Long Answer: Accept any relevant application of Newton's laws in engineering or mechanics with sufficient detail.

Rubric: Short Answer Questions: 1 point for a correct explanation, 1 point for clarity; Long Answer Questions:

3 points for understanding Newton's laws, 2 points for relevance to engineering or mechanics, 2 points for examples or depth of discussion.